



BULE HORA UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATICS
DEPARTMENT OF SOFTWARE ENGINEERING

Title: GLCAE Vision Hub: A Centralized Web Platform

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Abstract

This report presents the design and implementation of GLCAE Vision Hub, a centralized web platform developed during an industrial internship to address inefficient manual workflows and fragmented communication. Built with React, TypeScript, and Supabase, the system features secure authentication, role-based access, automated scheduling, and centralized dashboards. The platform improves efficiency, transparency, and overall operational performance.



Introduction

Guba Lafto Consulting Architects & Engineers PLC (GLCAE) is a consultancy firm providing architectural, engineering, and ICT services.



The organization previously relied on manual processes and email communication, leading to inefficiencies, data inconsistencies, and limited coordination. To address these challenges, a centralized web-based system was developed to automate recruitment, manage service requests, and provide role-based dashboards. The objective was to enhance operational efficiency, improve data security, and streamline departmental collaboration.

METHODOLOGY

DATA COLLECTION

- => Structured interviews with HR and administrative staff
- => Observation of existing workflows
- => Review of organizational documents

Development Approach

- => Agile and iterative development model
- => Continuous stakeholder feedback

Tools & Technologies

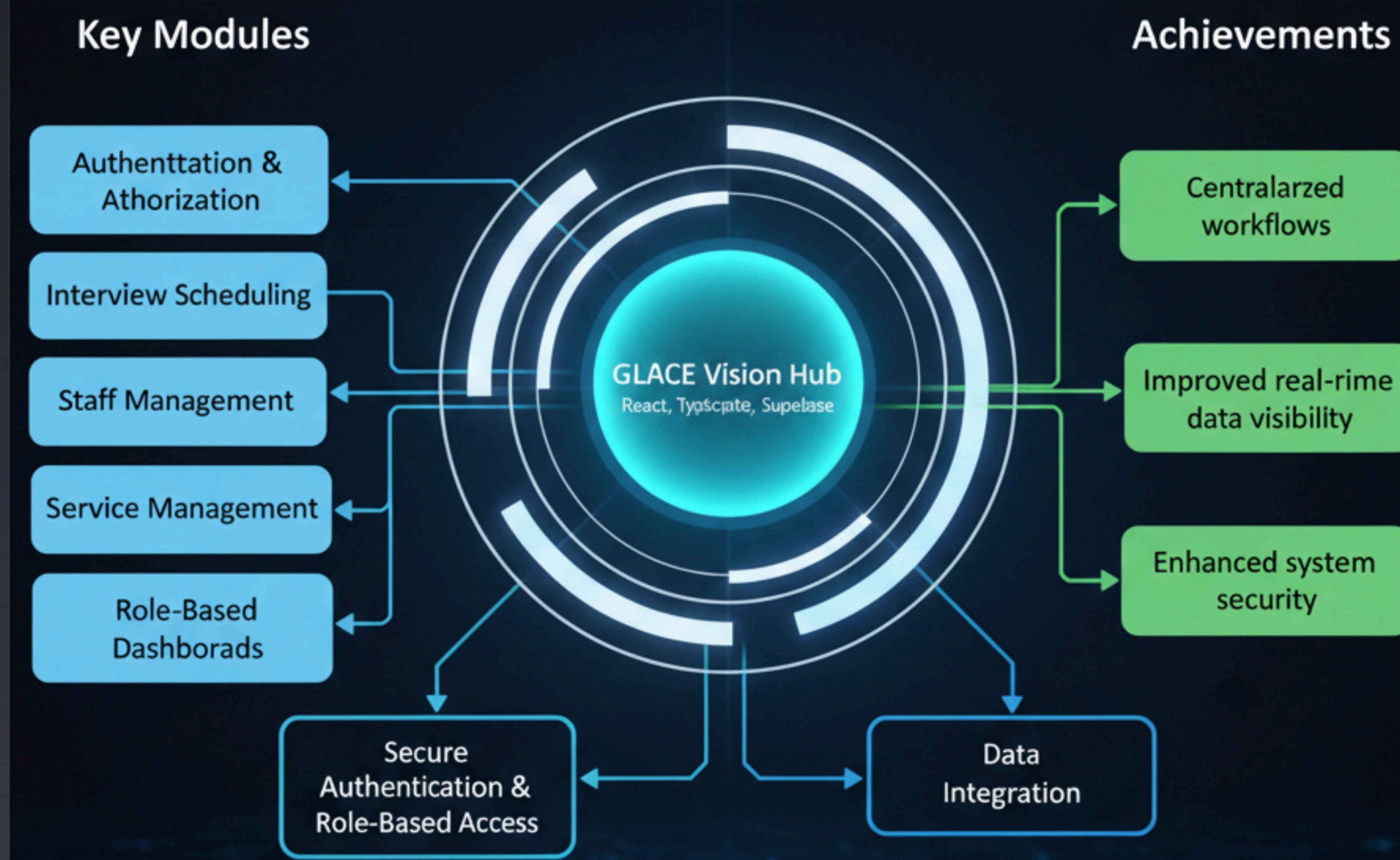
- => Frontend: React, TypeScript, Tailwind CSS
- => Backend: Supabase (PostgreSQL, Edge Functions, Storage)
- => Authentication: JWT-based system
- => Development Tools: VS Code, Git

Testing

- => Unit Testing
- => Integration Testing
- => User Acceptance Testing



IMPLEMENTATION & RESULTS



SECURITY FEATURES

- => Role-Based Access Control (RBAC)
- => JWT Authentication
- => Supabase Row-Level Security (RLS)
- => Input validation & data normalization
- => Protected routes in frontend

PROBLEM STATEMENT

- => Reliance on manual processes and spreadsheets.
- => Fragmented and uncoordinated workflows
- => Data duplication and inconsistencies
- => Interview scheduling conflicts
- => Limited transparency and reporting
- => Delayed communication between departments
- => Lack of a centralized digital system

SYSTEM ACHIEVEMENTS

- => Centralized all workflows into one platform
- => Reduced interview scheduling time significantly
- => Improved real-time data visibility
- => Minimized manual errors

Before

- => Manual scheduling
- => Data duplication
- => Limited reporting
- => Marketing Team

After

1. Centralized dashboard
2. Automated scheduling
3. Validated database
4. Instant visibility

CONCLUSION

GLCAE Vision Hub replaced manual workflows with a centralized system, improving efficiency and transparency using modern web technologies.